Genetic Algorithm

TSP & CartPole-v1

BIAI — Progress Report

Piotr Krupiński · Jeremi Szczotka

Politechnika Śląska · 2026

github.com/piotrek1459/biai-genetic-cartpole

Selected Topics

Task 1 — TSP

Traveling Salesman Problem

Find the shortest route visiting all cities exactly once and returning to the start.

- Classic combinatorial optimisation
- Chromosome: **permutation** of city indices
- Fitness: `1 / (distance + ε)`

Task 2 — CartPole-v1

Gymnasium reinforcement environment

Keep a pole balanced on a moving cart by pushing left or right.

- Control / policy learning task
- Chromosome: **real-valued** weight vector
- Fitness: mean episode reward

> Both problems are fundamentally different — TSP requires permutation operators, CartPole requires real-valued operators. One codebase handles both.

Project Goals

1. Implement a **generic GA framework** reusable across problem types
2. Implement all core **genetic operators** from scratch:
    - Selection: tournament, roulette
    - Crossover: OX, PMX (permutation) · uniform, arithmetic (real-valued)
    - Mutation: swap, inversion, scramble (permutation) · Gaussian (real-valued)
3. **Solve TSP** for 20 cities — minimise total route distance
4. **Solve CartPole** — learn a linear policy that achieves max reward (500)
5. **Compare hyperparameters** — operators, population size, mutation strength
6. Produce visualisations, comparison plots, and an agent video

Plan of Work

Phase	Task	Status
1	Repository scaffold, `requirements.txt`, virtualenv	✅ Done
2	Common operators — `selection.py`, `mutation.py`, `crossover.py`	✅ Done
3	Generic `GeneticAlgorithm` class (`ga_base.py`)	✅ Done
4	Task 1: TSP — training, visualisation, experiments	✅ Done
5	Task 2: CartPole — policy, training, agent video, experiments	✅ Done
6	README, `.gitignore`, documentation, progress report	✅ Done

All phases complete. Both algorithms run end-to-end and produce results.

Implementation Overview

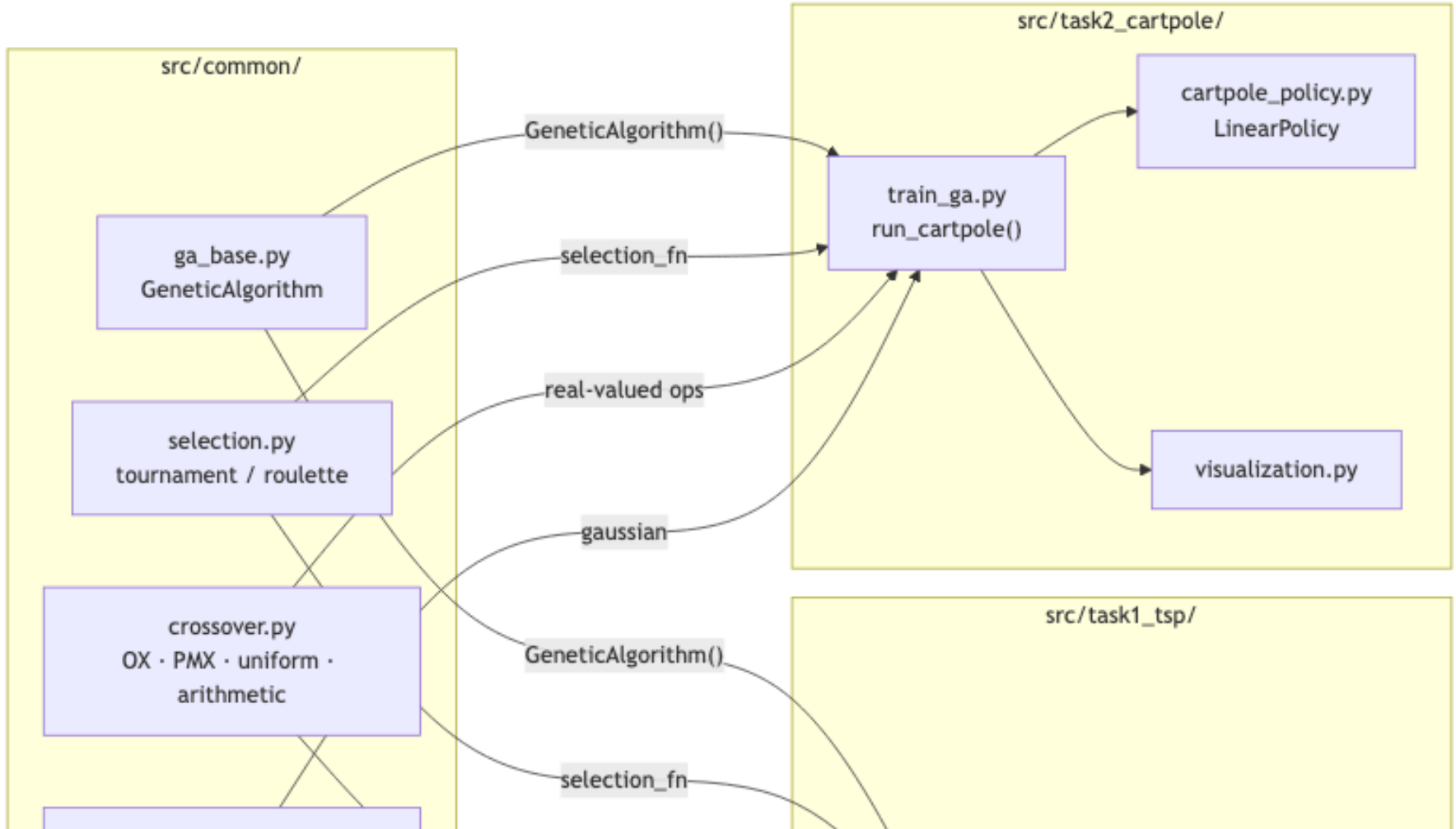
```
src/
  common/
    ga_base.py       ← GeneticAlgorithm class
    selection.py     ← tournament, roulette
    crossover.py     ← OX, PMX, uniform, arithmetic
    mutation.py      ← swap, inversion, scramble,
                        gaussian
  task1_tsp/
    tsp_ga.py        ← CONFIG + run_tsp()
    experiments.py   ← operator & population runs
  task2_cartpole/
    cartpole_policy.py  ← LinearPolicy
    train_ga.py         ← CONFIG + run_cartpole()
    evaluate.py         ← record agent as MP4
    experiments.py      ← sigma, pop, episodes
```

Design principle:

The `GeneticAlgorithm` class is **fully generic** — it receives operators as callables and

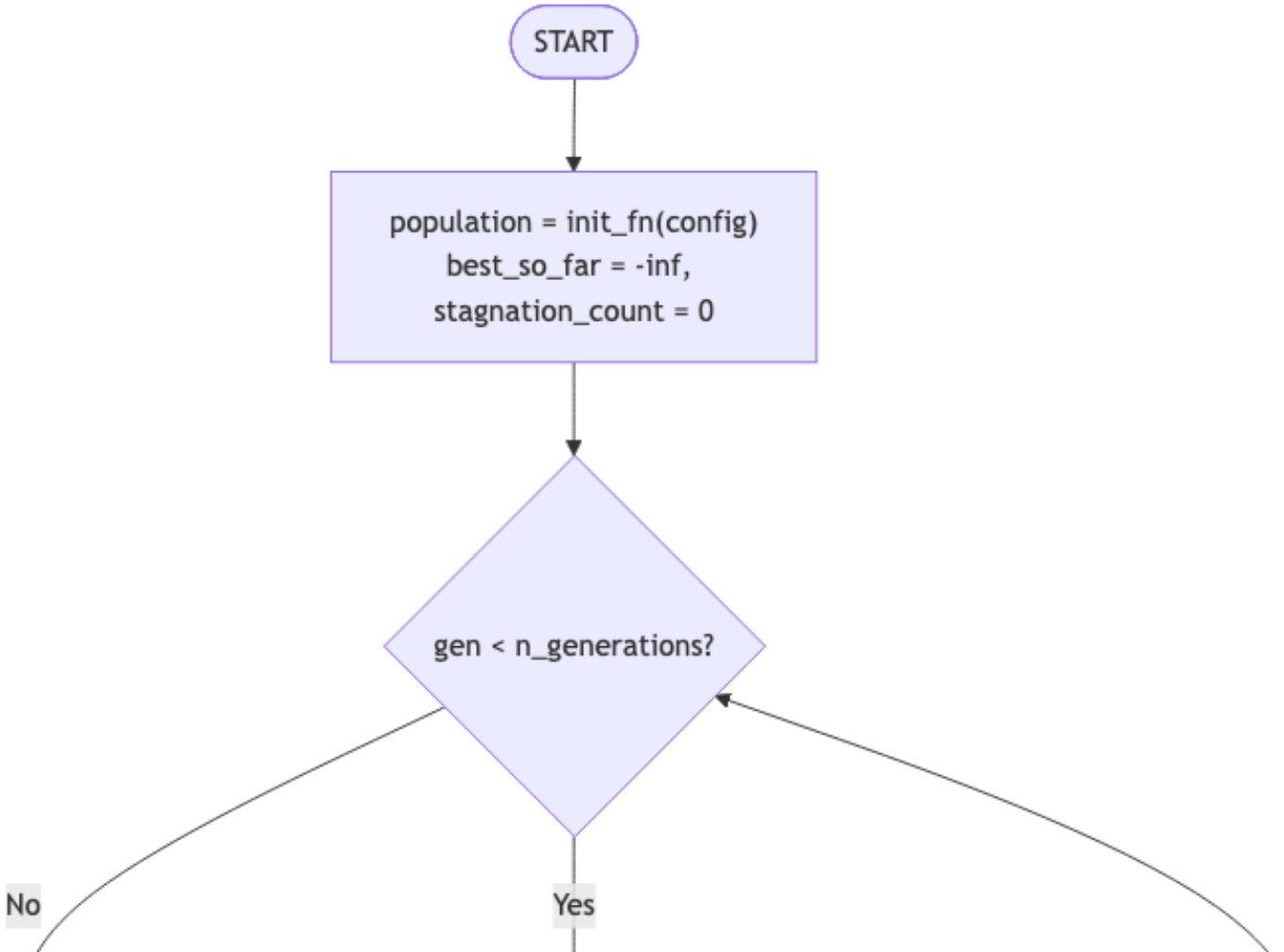
Module Architecture

How `src/common/` operators are shared between both tasks.



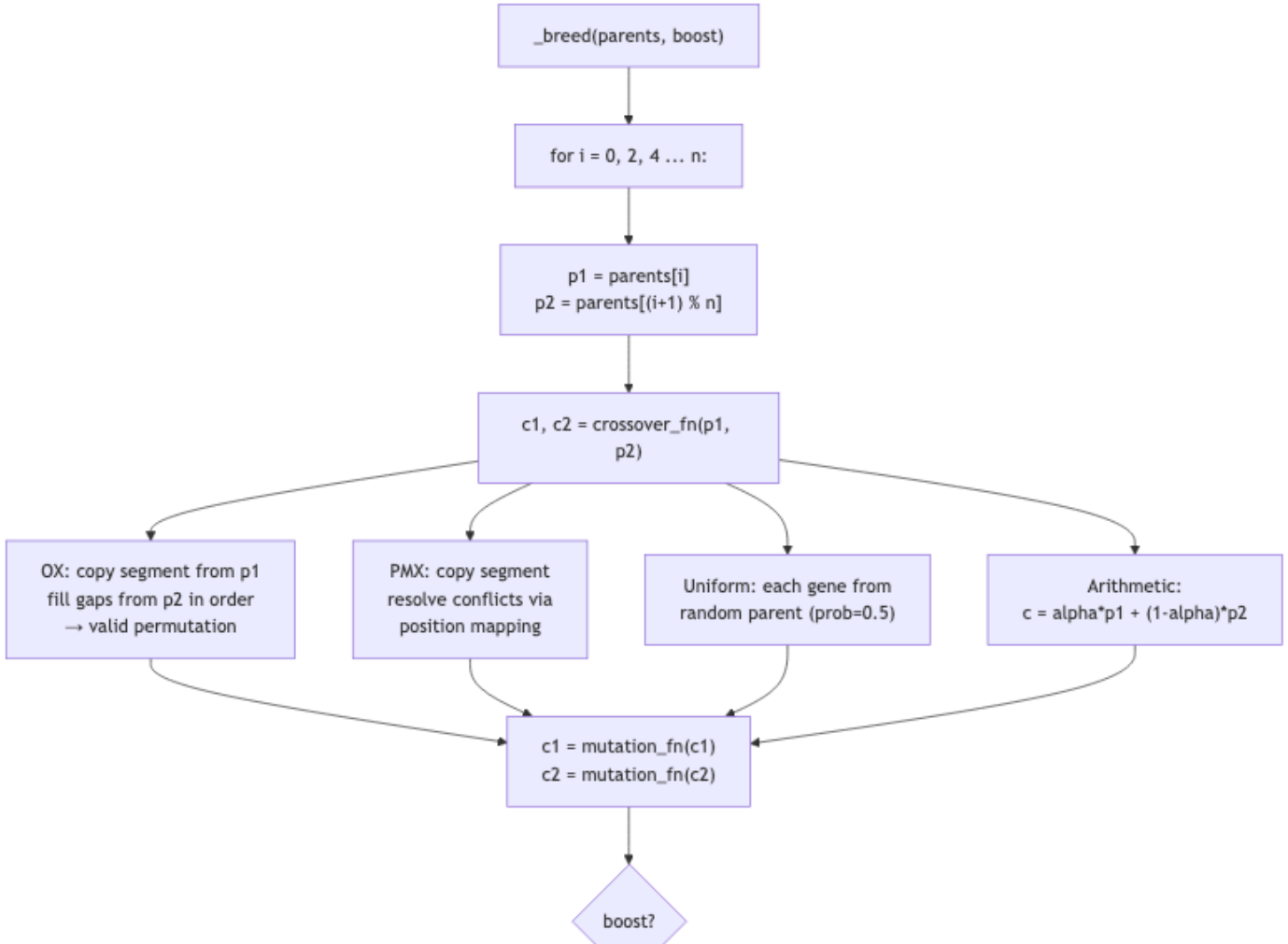
GA Loop — Core Algorithm (1/2)

The `GeneticAlgorithm.run()` loop — shared by both tasks.

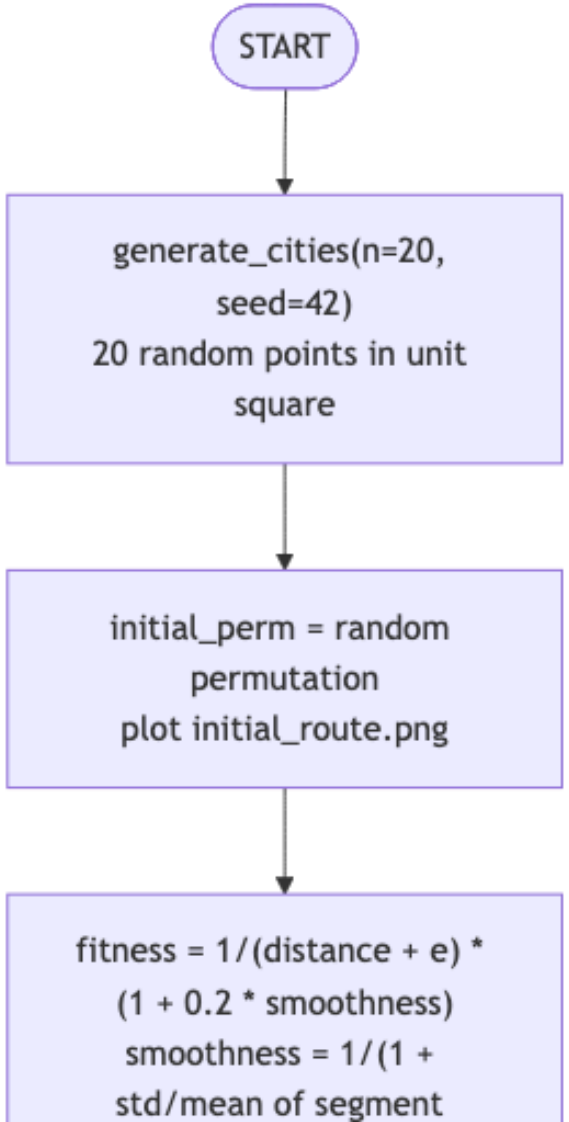


GA Loop — Core Algorithm (2/2): Breeding

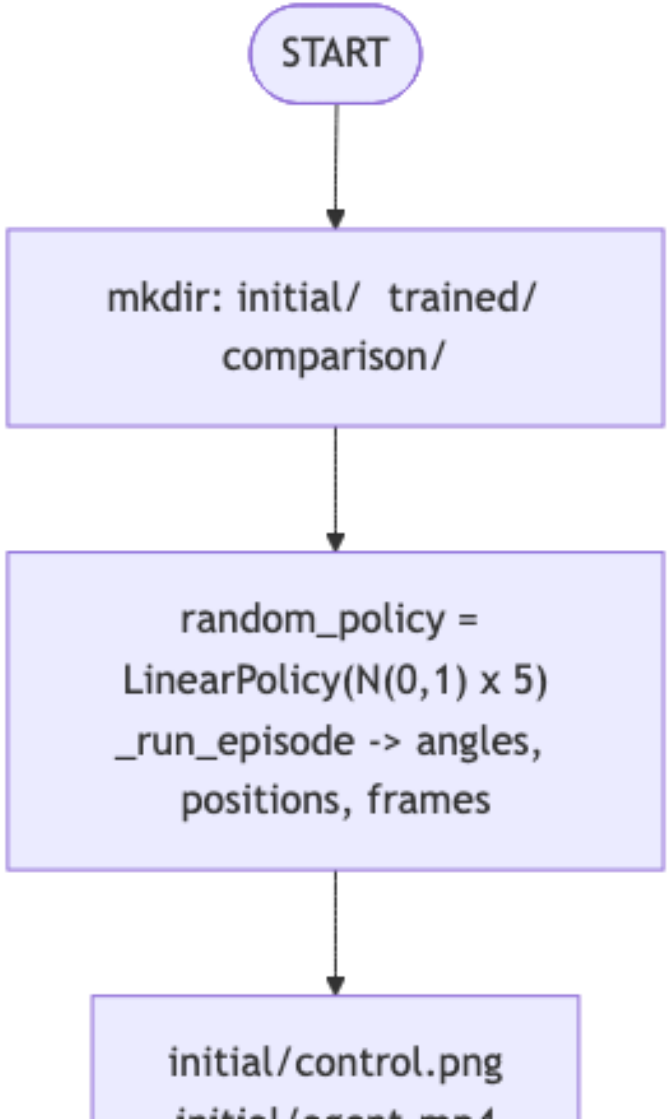
How each pair of parents produces offspring inside `_breed()`.



Task 1: TSP — Full Execution Flow



Task 2: CartPole — Full Execution Flow



GA Core — Genetic Operators

Selection

Method	Key property
Tournament (`k=4`)	Tunable pressure via `k`
Roulette wheel	Proportional to fitness

Crossover

Operator	Type
OX — Order Crossover	Permutation
PMX — Partially Mapped	Permutation
Uniform	Real-valued
Arithmetic blend	Real-valued

Mutation

Operator	Type
Swap	Permutation
Inversion *(default TSP)*	Permutation
Scramble	Permutation
Gaussian N(0, σ)	Real-valued

Elitism

Top `n_elite = 2` individuals are copied unchanged into the next generation, preventing loss of the best solution found so far.

Task 1: TSP — Algorithm

Chromosome: permutation of city indices, e.g. `[2, 0, 4, 1, 3]`
→ route: city 2 → city 0 → city 4 → city 1 → city 3 → city 2

Fitness:

$$f(\text{route}) = \frac{1}{\text{total distance} + \varepsilon}$$

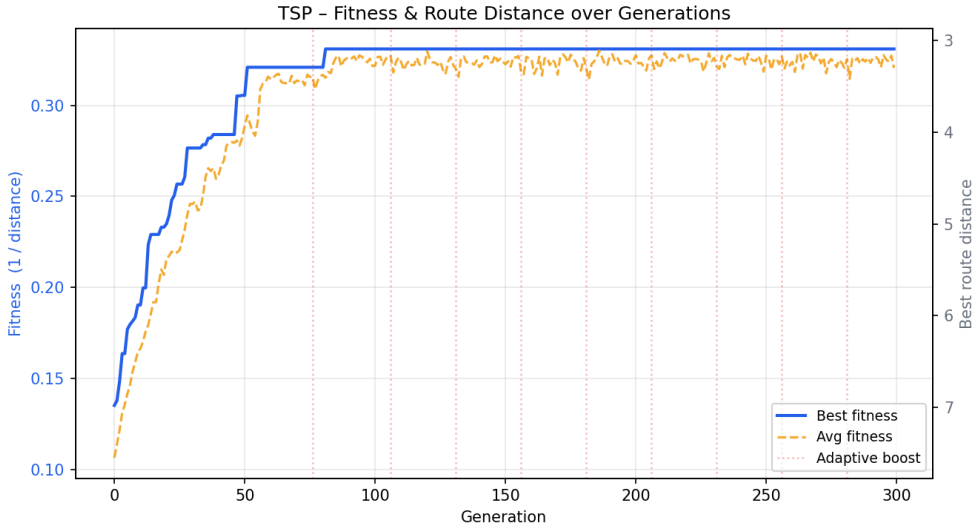
Maximising fitness ≡ minimising total Euclidean round-trip distance.

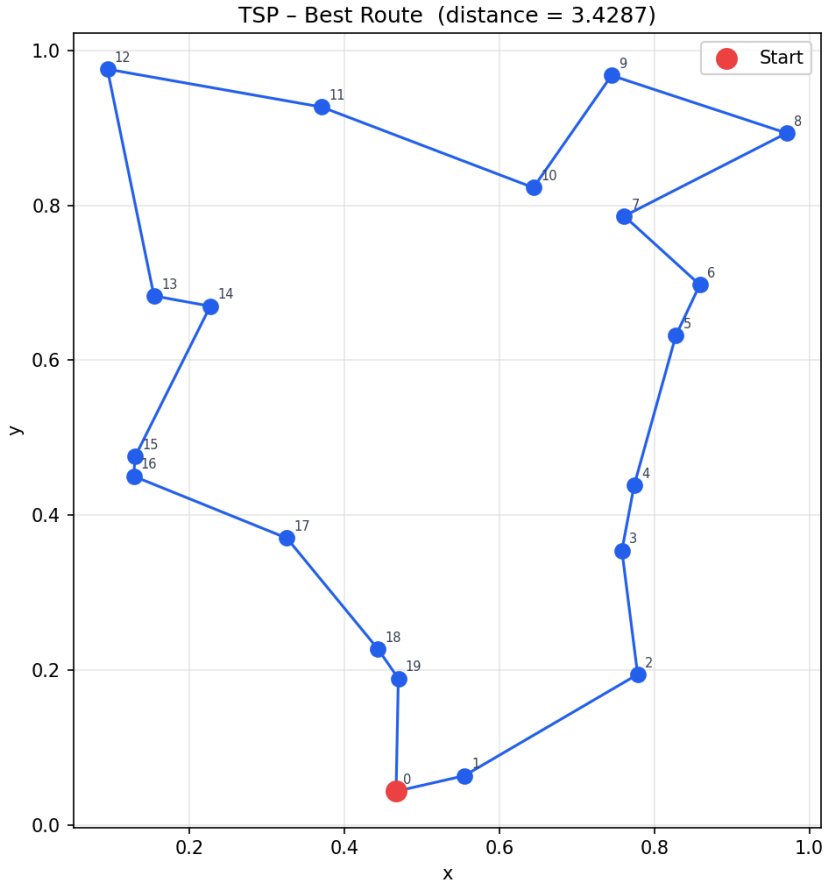
Default configuration:

pop	generations	crossover	mutation	elites	tournament k
80	300	OX	Inversion, rate=0.10	2	4

Cities: 20 random points in [0, 1]² · Theoretical optimum ≈ 3.1

Task 1: TSP — Results



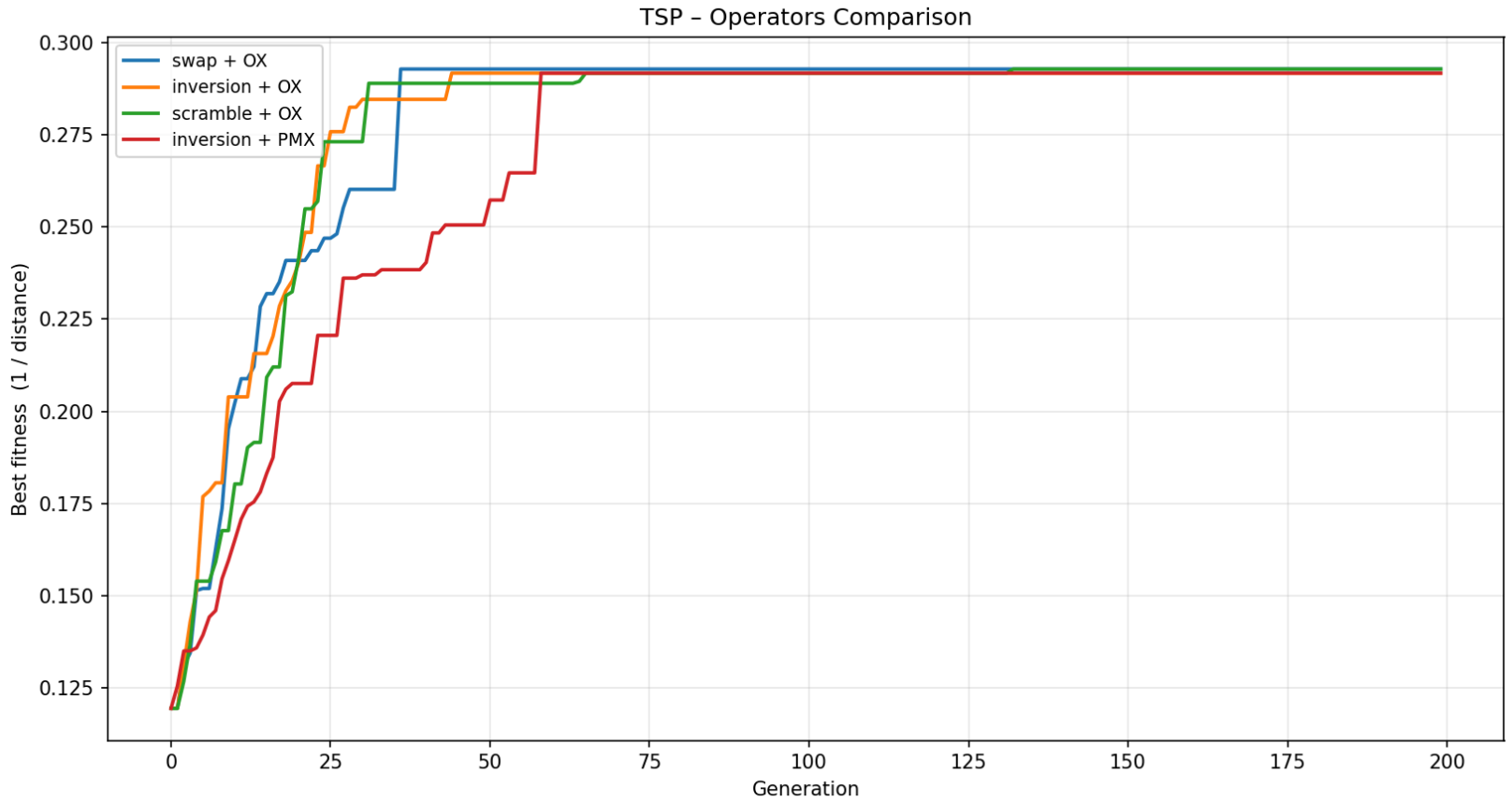


Fitness converges quickly in the first ~50 generations and stabilises around the near-optimal solution.

Best route distance: 3.43

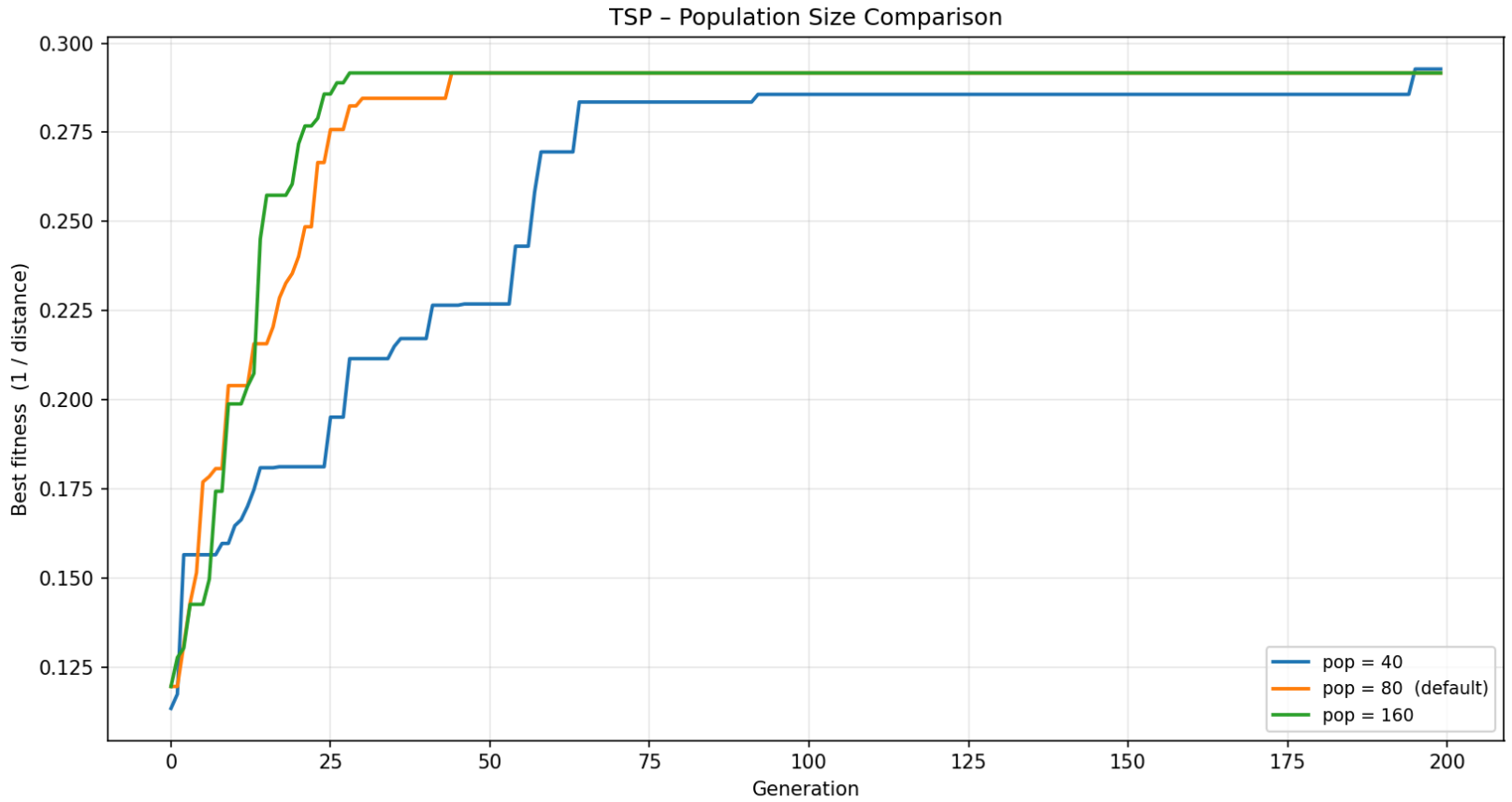
Theoretical optimum ≈ 3.1 — within ~10%

Task 1: TSP — Operator Comparison



All four operator combinations converge to a similar quality. **Inversion+OX** is the default choice — good balance of exploration and exploitation.

Task 1: TSP — Population Size Comparison



All population sizes reach similar final fitness. Larger populations show more stable convergence curves but require more computation per generation.

Task 2: CartPole — Algorithm

Environment: 4D observation `[cart_pos, cart_vel, pole_angle, pole_angular_vel]`

Action: push left (0) or push right (1) · **Max reward: 500**

Linear policy (chromosome = 5 real numbers):

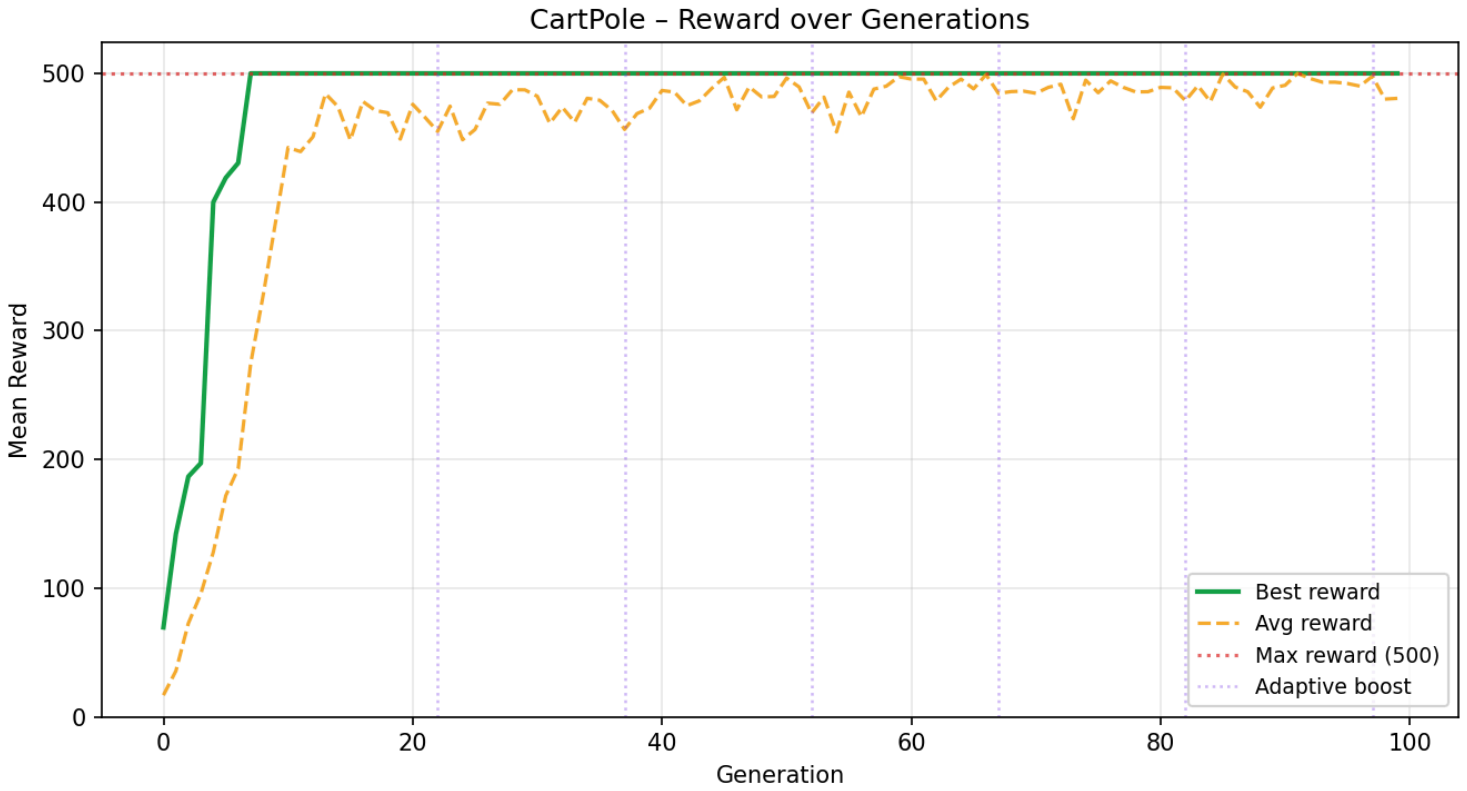
$$\text{action} = \begin{cases} 1 & \text{if } w_0 \cdot \text{pos} + w_1 \cdot \text{vel} + w_2 \cdot \theta + w_3 \cdot \dot{\theta} + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

Fitness: mean total reward over 5 independent episodes.

Default configuration:

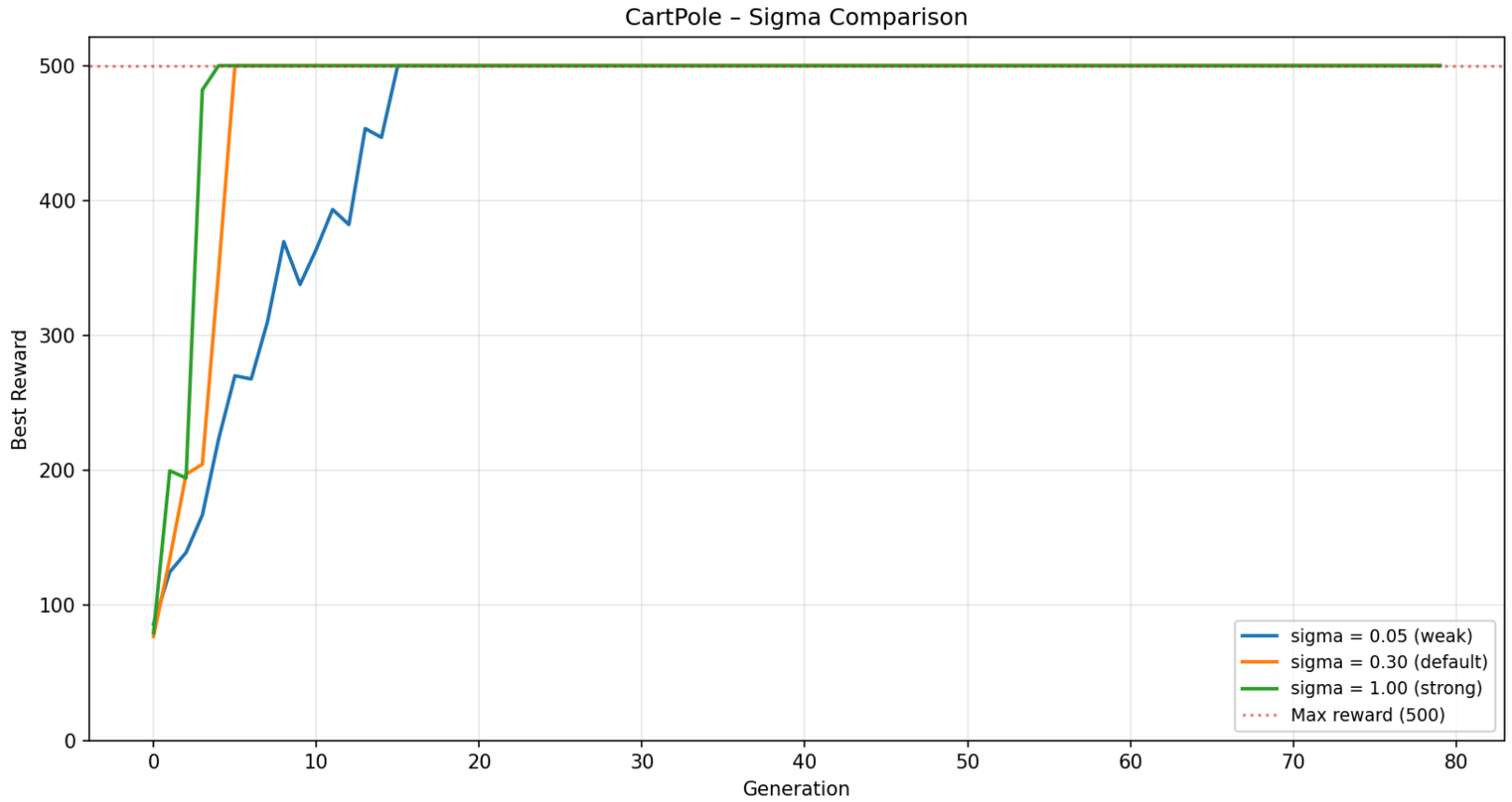
pop	generations	crossover	mutation	elites	episodes/eval
50	100	Uniform	Gaussian σ=0.3, rate=0.10	2	5

Task 2: CartPole — Results



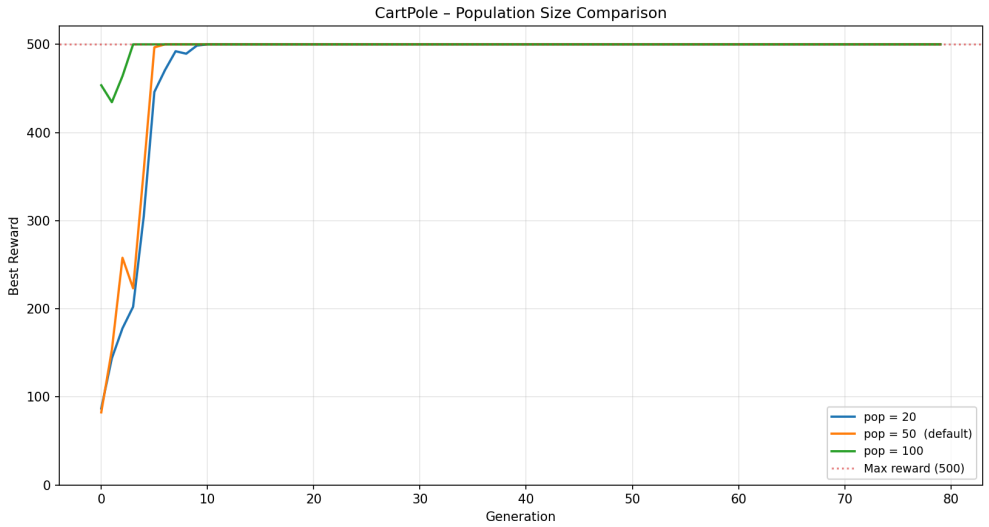
Maximum reward (500) reached at generation 10. The linear policy is sufficient to fully solve CartPole-v1. Agent video saved to `results/task2/agent.mp4`.

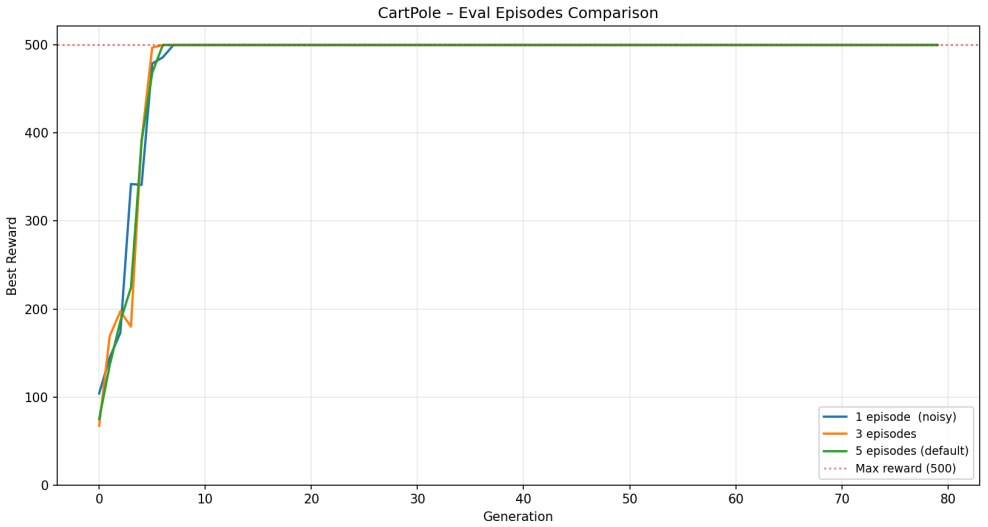
Task 2: CartPole — Sigma Comparison



All three sigma values eventually reach reward 500. Smaller σ converges more smoothly; larger σ shows higher variance but can escape local optima faster.

Task 2: CartPole — Population & Episodes Comparison





Larger populations start with better initial diversity — `pop=100` hits reward 500 within the first few generations.

More evaluation episodes reduce fitness noise, leading to smoother and more reliable convergence.

Conclusions

Task 1 — TSP ✅

- GA reduces route distance from ~6 (random) to **3.43** — within 10% of the theoretical optimum (~3.1)
- All operator combinations perform similarly; **OX + inversion** is the most consistent choice

Task 2 — CartPole ✅

- GA discovers a working **linear policy** in just **10 generations**
- Best agent achieves **reward = 500/500** every episode (maximum possible)
- CartPole-v1 is linearly solvable — a 5-parameter chromosome is sufficient

Key Takeaway

A single generic GA framework with **pluggable operators** handles both discrete (permutation) and continuous (real-valued) optimisation problems effectively.

Thank You

Repository: github.com/piotrek1459/biai-genetic-cartpole

```
# Run TSP
python3 -m src.task1_tsp.tsp_ga

# Run CartPole
python3 -m src.task2_cartpole.train_ga
python3 -m src.task2_cartpole.evaluate
```

Piotr Krupiński · Jeremi Szczotka